



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
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ASSIGNMENT No: 16

TITLE: Write a Java program to create and handle a user-defined exception.

Problem Statement

Write a Java program to create and handle a user-defined exception.

Learning Objectives

To create and implement custom exceptions for application-specific error handling.

Theory:

User-defined exceptions are created by extending the Exception class. They allow developers to define meaningful exceptions for specific business logic. Custom exceptions improve program readability and simplify debugging by providing descriptive error messages. They are widely used in enterprise software and validation systems.



Exercise

1. Create a Voting System that throws a custom exception if age is below 18

```
J VotingSystem.java > ...
1 package assignment16;
2 import java.util.Scanner;
3 public class DrivingLicenseSystem {
4     public static void verifyDrivingAge(int age) throws DrivingAgeException {
5         if (age < 18) {
6             throw new DrivingAgeException("Age is below 18. Application rejected.");
7         } else {
8             System.out.println("User is eligible for a driving license.");
9         }
10    }
11    Run main | Debug main
12    public static void main(String[] args) {
13        Scanner s = new Scanner(System.in);
14        System.out.print("Enter your age: ");
15        try {
16            verifyDrivingAge(s.nextInt());
17        } catch (DrivingAgeException e) {
18            System.out.println("Custom Exception Caught: " + e.getMessage());
19        } finally {
20            System.out.println("[INFO] Driving license verification process completed.");
21            s.close();
22        }
23    }
24    class DrivingAgeException extends Exception {
25        public DrivingAgeException(String msg) { super(msg); }
26    }
27 }
```

```
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 16> java VotingSystem.java
Enter your age: 17
Custom Exception Caught: Age is below 18. Application rejected.
[INFO] Driving license verification process completed.
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 16> java VotingSystem.java
Enter your age: 19
User is eligible for a driving license.
[INFO] Driving license verification process completed.
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 16> |
```



2. Create a Driving License System that throws a custom exception if the user's age is below 18. If the age is valid, display that the user is eligible for a driving license.

```
1 package assignment16;
2 import java.util.Scanner;
3 public class Drive {
4     public static void verifyDrivingAge(int age) throws DrivingAgeException {
5         if (age < 18) {
6             throw new DrivingAgeException("Age is below 18. Application rejected.");
7         } else {
8             System.out.println("User is eligible for a driving license.");
9         }
10    }
11    Run main | Debug main
12    public static void main(String[] args) {
13        Scanner s = new Scanner(System.in);
14        System.out.print("Enter your age: ");
15        try {
16            verifyDrivingAge(s.nextInt());
17        } catch (DrivingAgeException e) {
18            System.out.println("Custom Exception Caught: " + e.getMessage());
19        } finally {
20            System.out.println("[INFO] Driving license verification process completed.");
21            s.close();
22        }
23    }
24    class DrivingAgeException extends Exception {
25        public DrivingAgeException(String msg) { super(msg); }
26    }
27 }
```

```
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 16> javac Drive.java
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 16> java Drive.java
Enter your age: 17
Custom Exception Caught: Age is below 18. Application rejected.
[INFO] Driving license verification process completed.
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 16> java Drive.java
Enter your age: 19
User is eligible for a driving license.
[INFO] Driving license verification process completed.
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 16> |
```